

Project: Data Analytics on Global Tuberculosis Burden Across Countries.

Introduction:

Tuberculosis (TB) remains one of the top infectious diseases globally, despite being preventable and curable. According to the World Health Organization (WHO), millions of new cases occur every year, with varying prevalence across regions. Data analytics can reveal patterns in TB incidence, mortality, and treatment success rates—helping policymakers and health organizations better target interventions and allocate resources.

#	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	Country	iso2	iso3	iso_numeric	g_whoreg	year	e_pop_nune	e_inc_100k	e_inc_100k	e_inc_100k	e_inc_num	e_inc_num	e_inc_num	e_inc_num	e_tbhiv_prie	e_tbhiv_prie	e_tbhiv_prie	hive
2	Afghanistan	AF	AFG	4	EMR	2000	20130323	190	122	271	38000	25000	55000	0.02	0	0	Exit Full Screen	0
3	Afghanistan	AF	AFG	4	EMR	2001	20284311	189	122	271	38000	25000	55000	0.01	0	0.04	0.03	0
4	Afghanistan	AF	AFG	4	EMR	2002	21378110	189	122	270	40000	26000	58000	0.01	0	0.04	0.03	0
5	Afghanistan	AF	AFG	4	EMR	2003	22733047	189	122	270	43000	28000	61000	0.01	0	0.04	0.03	0
6	Afghanistan	AF	AFG	4	EMR	2004	23560660	189	122	270	44000	29000	64000	0.01	0	0.04	0.03	0
7	Afghanistan	AF	AFG	4	EMR	2005	24404569	189	122	270	46000	30000	66000	0.02	0	0.05	0.03	0
8	Afghanistan	AF	AFG	4	EMR	2006	25424099	189	122	270	48000	31000	69000	0.02	0	0.05	0.03	0
9	Afghanistan	AF	AFG	4	EMR	2007	25909852	189	122	270	49000	32000	70000	0.02	0	0.06	0.04	0
10	Afghanistan	AF	AFG	4	EMR	2008	26482615	189	122	271	50000	32000	72000	0.02	0	0.06	0.04	0
11	Afghanistan	AF	AFG	4	EMR	2009	27466100	189	122	271	52000	34000	74000	0.02	0	0.07	0.04	0
12	Afghanistan	AF	AFG	4	EMR	2010	28284084	189	122	271	54000	35000	77000	0.02	0	0.07	0.05	0
13	Afghanistan	AF	AFG	4	EMR	2011	29347711	189	122	271	56000	36000	79000	0.03	0	0.08	0.05	0
14	Afghanistan	AF	AFG	4	EMR	2012	30560030	189	122	270	58000	37000	83000	0.03	0	0.08	0.05	0
15	Afghanistan	AF	AFG	4	EMR	2013	31622705	189	122	270	60000	39000	85000	0.03	0	0.09	0.05	0
16	Afghanistan	AF	AFG	4	EMR	2014	32792530	189	122	270	62000	40000	88000	0.03	0	0.09	0.05	0
17	Afghanistan	AF	AFG	4	EMR	2015	33831765	189	122	270	64000	41000	91000	0.03	0	0.09	0.05	0
18	Afghanistan	AF	AFG	4	EMR	2016	34700602	189	122	270	65000	42000	94000	0.03	0	0.09	0.05	0
19	Afghanistan	AF	AFG	4	EMR	2017	35688933	189	122	270	67000	44000	96000	0.03	0	0.1	0.05	0
20	Afghanistan	AF	AFG	4	EMR	2018	36743039	189	122	270	69000	45000	99000	0.03	0	0.1	0.05	0
21	Afghanistan	AF	AFG	4	EMR	2019	37856118	189	122	270	71000	46000	102000	0.03	0	0.1	0.05	0
22	Afghanistan	AF	AFG	4	EMR	2020	39068984	183	118	262	72000	46000	102000	0.02	0.01	0.03	0.03	0.01
23	Afghanistan	AF	AFG	4	EMR	2021	40000415	185	117	264	74000	47000	106000	0.01	0	0.02	0.01	0
24	Afghanistan	AF	AFG	4	EMR	2022	40578848	184	115	265	75000	47000	108000	0.03	0.02	0.05	0.06	0.03
25	Afghanistan	AF	AFG	4	EMR	2023	41454760	180	112	263	75000	47000	109000	0.03	0.01	0.04	0.05	0.02
26	Albania	AL	ALB	8	EUR	2000	3166141	22	18	27	700	570	850	0.15	0.05	0.3	0.03	0.01
27	Albania	AL	ALB	8	EUR	2001	3147944	20	17	24	640	530	770	0.14	0.05	0.28	0.03	0.01
28	Albania	AL	ALB	8	EUR	2002	3134100	22	18	26	680	560	830	0.15	0.05	0.3	0.03	0.01
29	Albania	AL	ALB	8	EUR	2003	3118092	20	17	24	640	520	760	0.18	0.06	0.35	0.04	0.01
30	Albania	AL	ALB	8	EUR	2004	3098661	20	17	24	630	520	750	0.22	0.08	0.42	0.04	0.02
31	Albania	AL	ALB	8	EUR	2005	3076163	19	16	22	580	480	680	0.27	0.1	0.54	0.05	0.02
32	Albania	AL	ALB	8	EUR	2006	3050802	18	15	21	540	450	630	0.34	0.12	0.67	0.06	0.02

● To analyze the country-wise TB burden using WHO datasets.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv('/content/TB_burden_countries_2025-10-31.csv')

latest_year = df['year'].max()
df_latest = df[df['year'] == latest_year].copy()

df_latest = df_latest.dropna(subset=['e_inc_100k', 'e_mort_100k'])

country_burden = df_latest.groupby('country').agg({
    'e_inc_100k': 'mean',
    'e_mort_100k': 'mean'
}).reset_index()

top_countries = country_burden.sort_values(by='e_inc_100k', ascending=False).head(10)

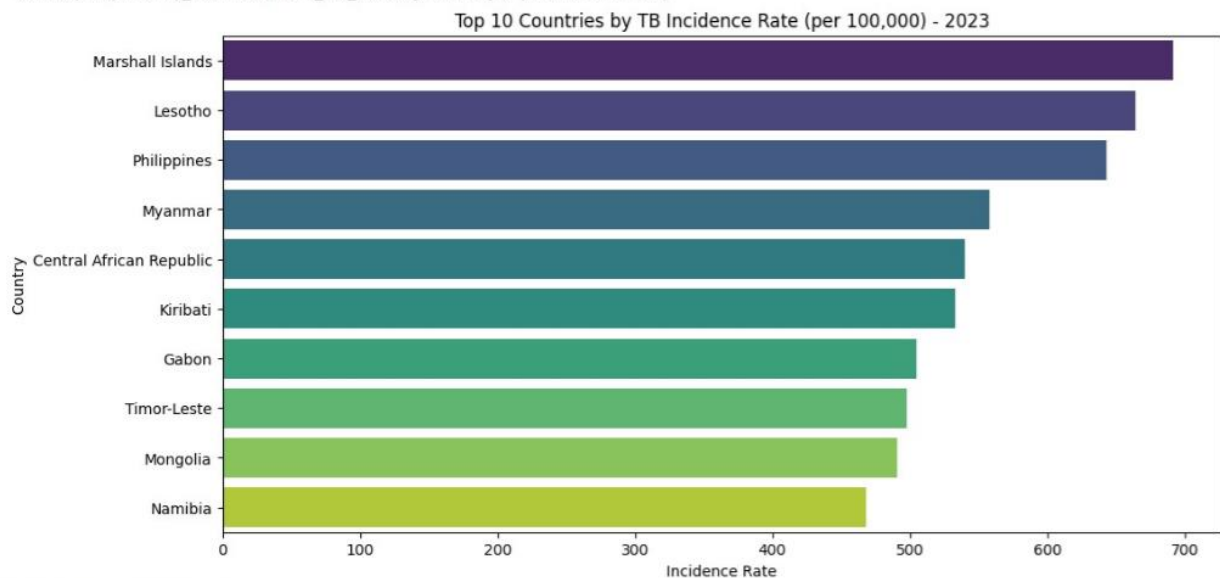
plt.figure(figsize=(12, 6))
sns.barplot(data=top_countries, x='e_inc_100k', y='country', palette='viridis')
plt.title(f'Top 10 Countries by TB Incidence Rate (per 100,000) - {latest_year}')
plt.xlabel('Incidence Rate')
plt.ylabel('Country')
plt.show()

print("Summary of TB Burden Analysis:")
print(f"Total countries analyzed: {len(country_burden)}")
print(f"Global average incidence: {country_burden['e_inc_100k'].mean():.2f} per 100k")
print(f"Top country by incidence: {top_countries.iloc[0]['country']} ({top_countries.iloc[0]['e_inc_100k']:.2f} per 100k)")
```

`/tmp/ipython-input-3082011014.py:44: FutureWarning:`

• Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for

```
sns.barplot(data=top_countries, x='e_inc_100k', y='country', palette='viridis')
```



Summary of TB Burden Analysis:

- To study trends in TB incidence, prevalence, and mortality over time.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Step 1: Load the TB dataset (choose CSV or Excel; replace with your file path)
# For CSV: df = pd.read_csv('tb_data.csv')
# For Excel: df = pd.read_excel('tb_data.xlsx', sheet_name='Sheet1')
df = pd.read_csv('/content/TB_burden_countries_2025-10-31.csv') # Example: WHO TB burden data

# Step 2: Data cleaning
# Ensure 'year' is numeric and drop missing values for key metrics
df['year'] = pd.to_numeric(df['year'], errors='coerce')
# Removed 'e_prev_100k' from dropna subset as it does not exist
df = df.dropna(subset=['year', 'e_inc_100k', 'e_mort_100k'])

# Step 3: Aggregate trends globally by year
# Calculate mean values per year for global trends
yearly_trends = df.groupby('year').agg({
    'e_inc_100k': 'mean', # Average incidence per 100k
    # Removed 'e_prev_100k' as it caused KeyError
    'e_mort_100k': 'mean' # Average mortality per 100k
}).reset_index()

# Step 4: Visualization - Line plots for trends over time
plt.figure(figsize=(14, 8))

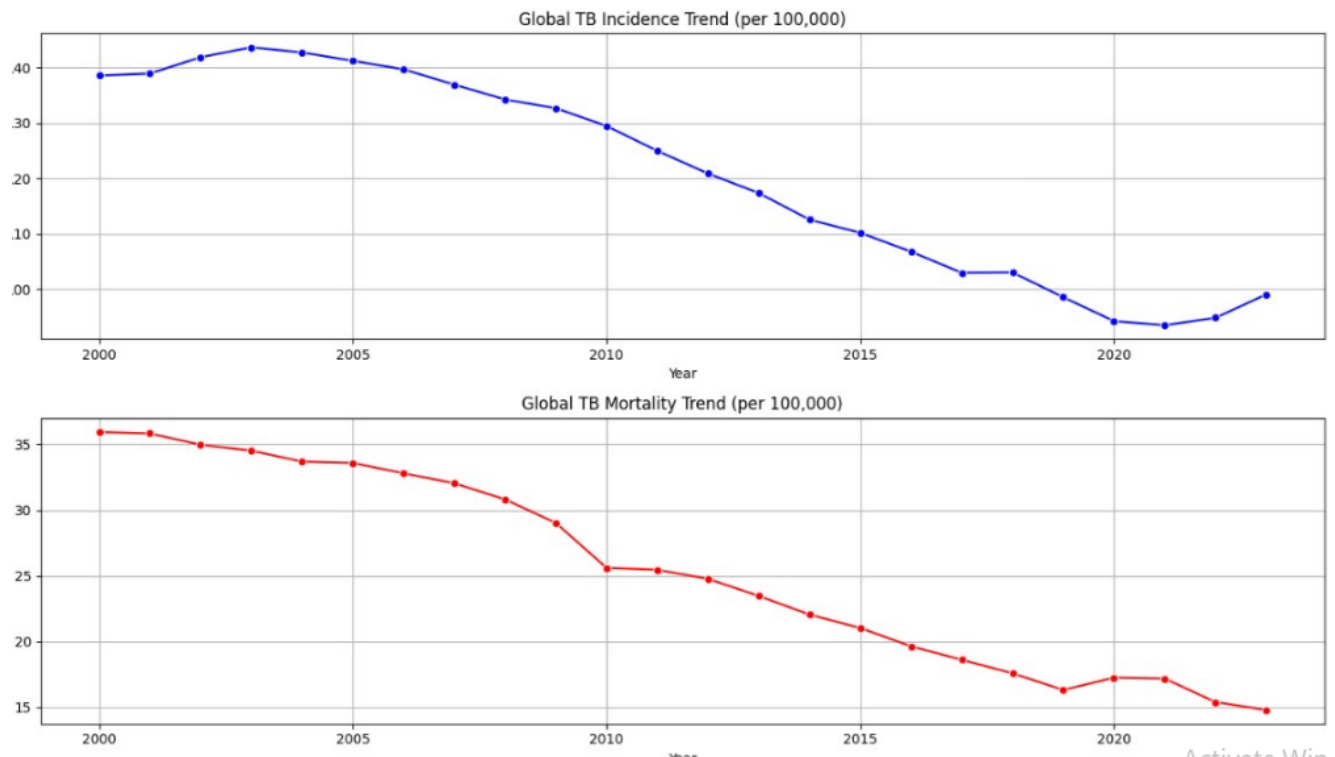
# Subplot 1: Incidence
plt.subplot(2, 1, 1) # Changed from 3,1,1 to 2,1,1
sns.lineplot(data=yearly_trends, x='year', y='e_inc_100k', marker='o', color='blue')
plt.title('Global TB Incidence Trend (per 100,000)')
plt.xlabel('Year')
plt.ylabel('Incidence Rate')
plt.grid(True)

# Removed Subplot 1: Prevalence as e_prev_100k column does not exist

# Subplot 2: Mortality
plt.subplot(2, 1, 2) # Changed from 3,1,3 to 2,1,2
sns.lineplot(data=yearly_trends, x='year', y='e_mort_100k', marker='o', color='red')
plt.title('Global TB Mortality Trend (per 100,000)')
plt.xlabel('Year')
plt.ylabel('Mortality Rate')
plt.grid(True)

plt.tight_layout()
plt.show()

# Optional: Print summary of trends
print("Summary of TB Trends Over Time:")
print(f"Years analyzed: {yearly_trends['year'].min()} to {yearly_trends['year'].max()}")
print(f"Incidence trend: Started at {yearly_trends.iloc[0]['e_inc_100k']:.2f}, ended at {yearly_trends.iloc[-1]['e_inc_100k']:.2f} per 100k")
# Removed Prevalence trend summary
print(f"Mortality trend: Started at {yearly_trends.iloc[0]['e_mort_100k']:.2f}, ended at {yearly_trends.iloc[-1]['e_mort_100k']:.2f} per 100k")
```



Summary of TB Trends Over Time:

Years analyzed: 2000 to 2023

Incidence trend: Started at 138.58, ended at 99.07 per 100k

Mortality trend: Started at 35.93, ended at 14.79 per 100k

- To compare TB cases across regions (Africa, Asia, Europe, etc.).

```

import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Step 1: Load the TB dataset (replace with your file path; CSV or Excel)
df = pd.read_csv('/content/TB_burden_countries_2025-10-31.csv') # Or pd.read_excel('tb_data.xlsx', sheet_name='Sheet1')

# Step 2: Data cleaning and region mapping
# If 'region' column exists, use it; otherwise, map countries to regions
if 'region' not in df.columns:
    # Simple mapping dictionary (expand as needed for full coverage)
    region_map = {
        'Afghanistan': 'Asia', 'Albania': 'Europe', 'Algeria': 'Africa', 'Angola': 'Africa',
        'Argentina': 'Americas', 'Australia': 'Western Pacific', 'Bangladesh': 'Asia',
        'Brazil': 'Americas', 'Canada': 'Americas', 'China': 'Western Pacific', 'Egypt': 'Africa',
        'France': 'Europe', 'Germany': 'Europe', 'India': 'Asia', 'Indonesia': 'Asia',
        'Italy': 'Europe', 'Japan': 'Western Pacific', 'Kenya': 'Africa', 'Mexico': 'Americas',
        'Nigeria': 'Africa', 'Pakistan': 'Asia', 'Russia': 'Europe', 'South Africa': 'Africa',
        'South Korea': 'Western Pacific', 'Spain': 'Europe', 'Thailand': 'Asia', 'Turkey': 'Europe',
        'United Kingdom': 'Europe', 'United States': 'Americas', 'Vietnam': 'Western Pacific',
        # Add more countries as needed
    }
    df['region'] = df['country'].map(region_map).fillna('Other') # Assign 'Other' for unmapped

# Filter for the latest year and drop missing values
latest_year = df['year'].max()
df_latest = df[df['year'] == latest_year].dropna(subset=['e_inc_100k', 'e_mort_100k'])

# Step 3: Aggregate by region
regional_burden = df_latest.groupby('region').agg({
    'e_inc_100k': 'mean', # Average incidence per 100k
    'e_mort_100k': 'mean' # Average mortality per 100k
}).reset_index()

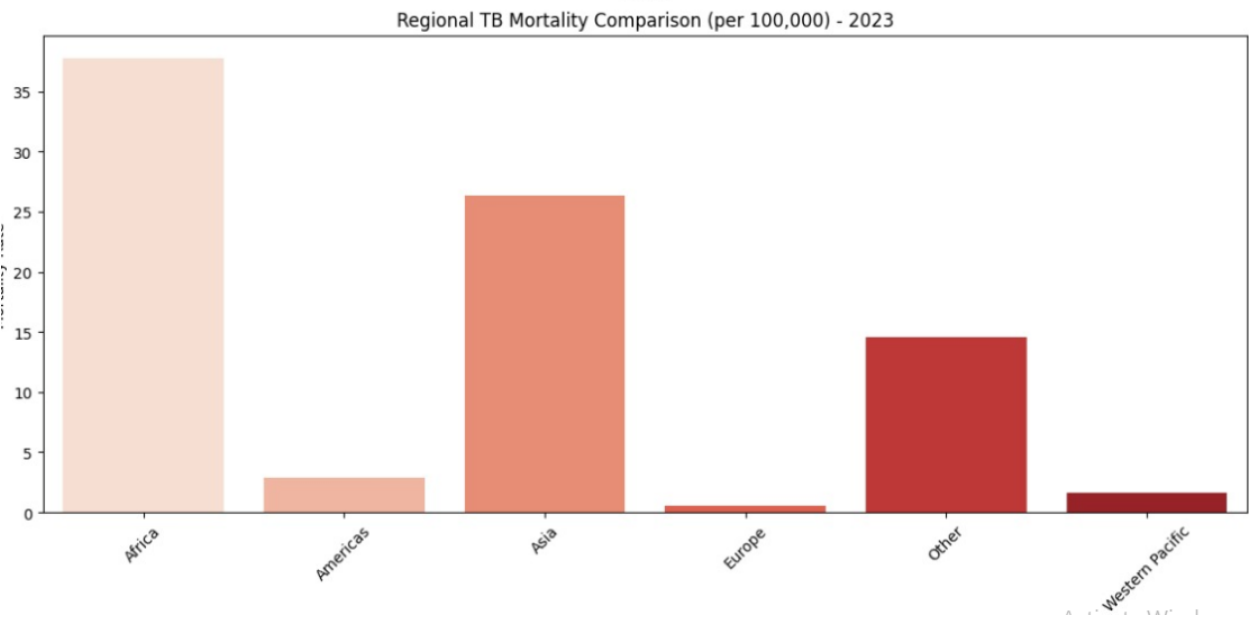
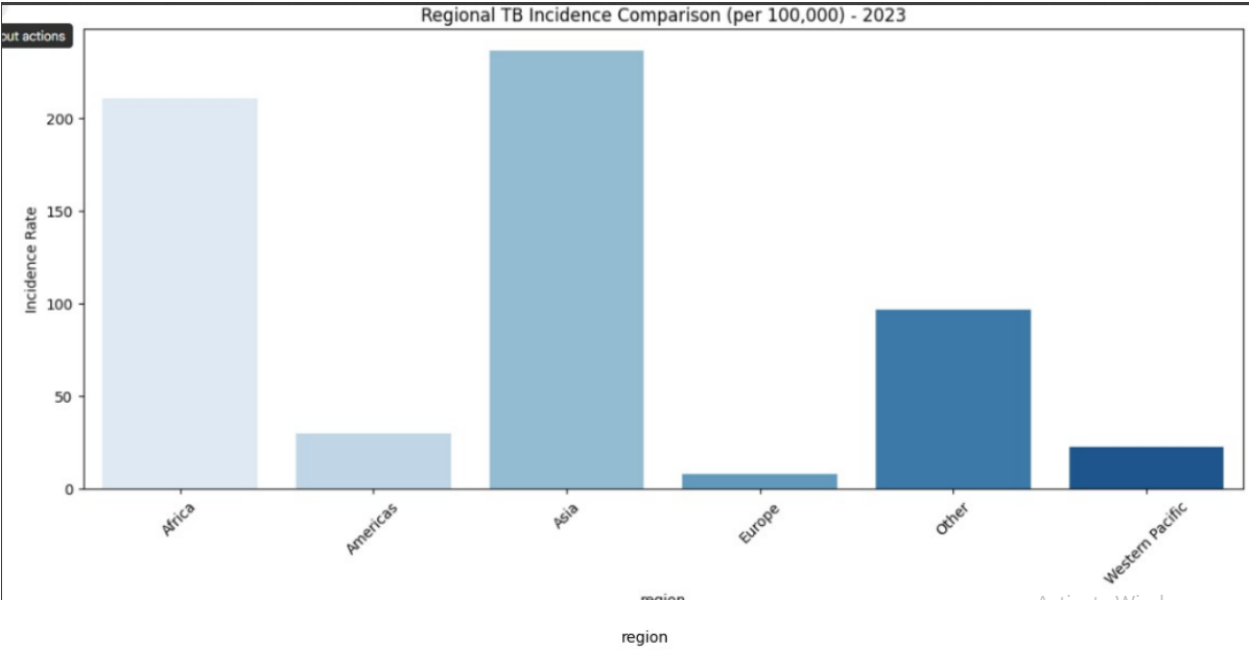
```

/tmp/ipython-input-3102035619.py:39: FutureWarning:

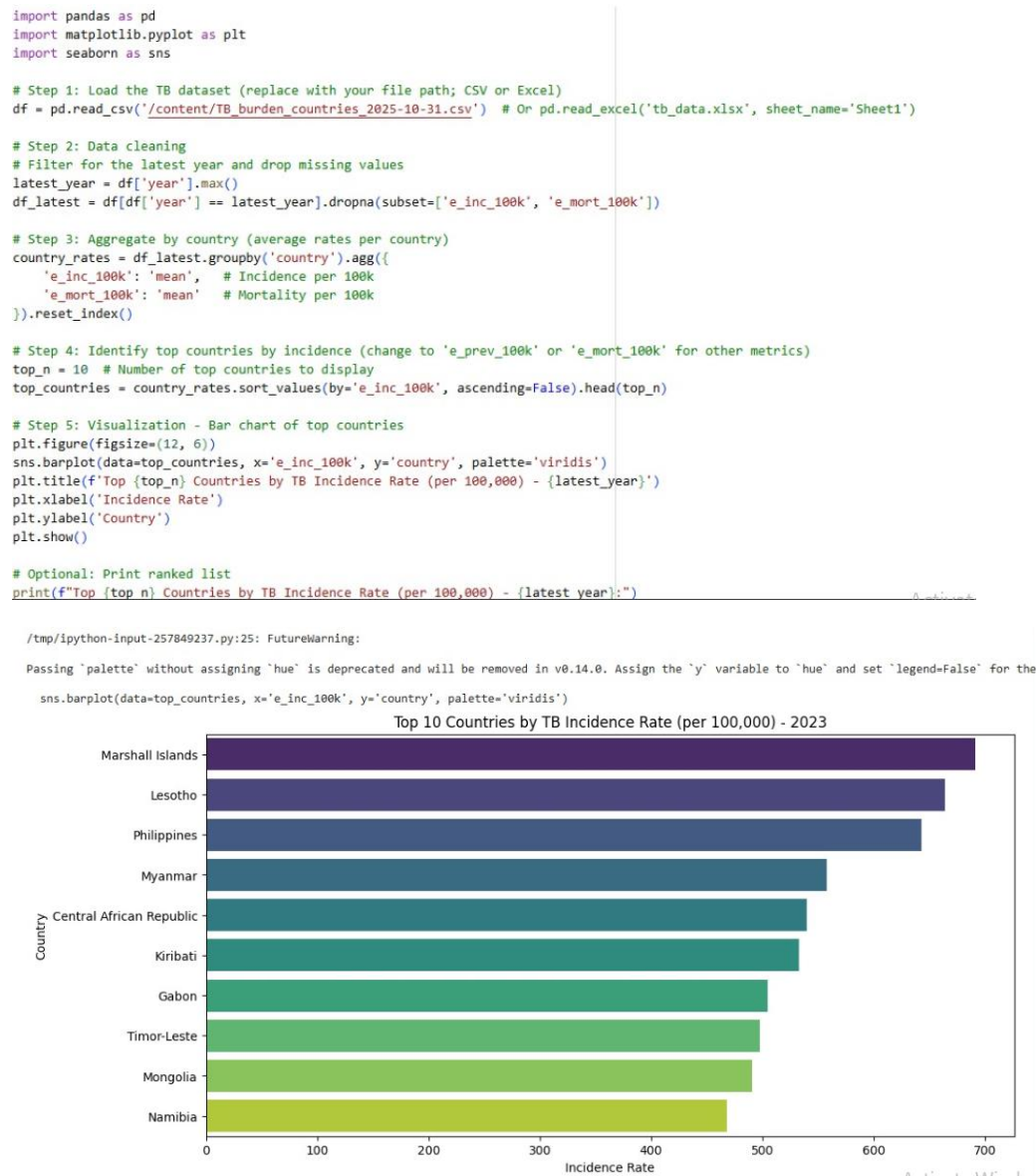
passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for th
 sns.barplot(data=regional_burden, x='region', y='e_inc_100k', ax=axes[0], palette='Blues')

/tmp/ipython-input-3102035619.py:45: FutureWarning:

passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for th
 sns.barplot(data=regional_burden, x='region', y='e_mort_100k', ax=axes[1], palette='Reds')



● To identify countries with the highest TB rates per 100,000 population.



Top 10 Countries by TB Incidence Rate (per 100,000) - 2023:

119. Marshall Islands: 692.00 per 100k (Mortality: 76.00)

108. Lesotho: 664.00 per 100k (Mortality: 229.00)

151. Philippines: 643.00 per 100k (Mortality: 32.00)

130. Myanmar: 558.00 per 100k (Mortality: 90.00)

39. Central African Republic: 540.00 per 100k (Mortality: 142.00)

102. Kiribati: 533.00 per 100k (Mortality: 37.00)

73. Gabon: 505.00 per 100k (Mortality: 139.00)

189. Timor-Leste: 498.00 per 100k (Mortality: 40.00)

125. Mongolia: 491.00 per 100k (Mortality: 12.00)

131. Namibia: 468.00 per 100k (Mortality: 98.00)

- To explore correlations between TB burden and socioeconomic indicators (like GDP, healthcare spending).

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np

# Step 1: Load TB dataset (replace with your file path)
tb_df = pd.read_csv('/content/TB_burden_countries_2025-10-31.csv') # WHO TB data with 'country', 'year', 'e_inc_100k', etc.

# Step 2: Define socio_df (FIX: Added placeholder socio_df as it was undefined)
# This is a placeholder. In a real scenario, you would load this from a file.
socio_data = {
    'country': ['Afghanistan', 'Albania', 'Algeria', 'United States', 'India', 'China', 'South Africa'],
    'gdp_per_capita': [500, 5000, 4000, 65000, 2000, 10000, 6000],
    'health_spending_per_capita': [50, 400, 300, 10000, 100, 800, 500]
}
socio_df = pd.DataFrame(socio_data)

# Step 3: Data cleaning and merging
# Filter TB data for the latest year
latest_year = tb_df['year'].max()
tb_latest = tb_df[tb_df['year'] == latest_year].dropna(subset=['e_inc_100k'])

# Merge on 'country' (ensure country names match; use fuzzy matching if needed)
merged_df = pd.merge(tb_latest, socio_df, on='country', how='inner')

# Drop rows with missing socioeconomic data
merged_df = merged_df.dropna(subset=['gdp_per_capita', 'health_spending_per_capita', 'e_inc_100k'])

# Step 4: Calculate correlations
# Pearson correlation matrix for key variables
correlation_matrix = merged_df[['e_inc_100k', 'gdp_per_capita', 'health_spending_per_capita']].corr()
print("Correlation Matrix:")
print(correlation_matrix)

# Step 5: Visualization
fig, axes = plt.subplots(2, 2, figsize=(14, 10))

# Scatter plot: TB Incidence vs. GDP
sns.scatterplot(data=merged_df, x='gdp_per_capita', y='e_inc_100k', ax=axes[0, 0])
axes[0, 0].set_title('TB Incidence vs. GDP per Capita')
axes[0, 0].set_xlabel('GDP per Capita (USD)')
axes[0, 0].set_ylabel('TB Incidence (per 100,000)')

# Scatter plot: TB Incidence vs. Healthcare Spending
sns.scatterplot(data=merged_df, x='health_spending_per_capita', y='e_inc_100k', ax=axes[0, 1])
axes[0, 1].set_title('TB Incidence vs. Healthcare Spending per Capita')
axes[0, 1].set_xlabel('Healthcare Spending per Capita (USD)')
axes[0, 1].set_ylabel('TB Incidence (per 100,000)')

# Correlation heatmap
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', ax=axes[1, 0])
axes[1, 0].set_title('Correlation Heatmap')

# Optional: Regression line on one plot (e.g., GDP)
sns.regplot(data=merged_df, x='gdp_per_capita', y='e_inc_100k', ax=axes[1, 1])
axes[1, 1].set_title('TB Incidence vs. GDP (with Regression Line)')
axes[1, 1].set_xlabel('GDP per Capita (USD)')
axes[1, 1].set_ylabel('TB Incidence (per 100,000)')

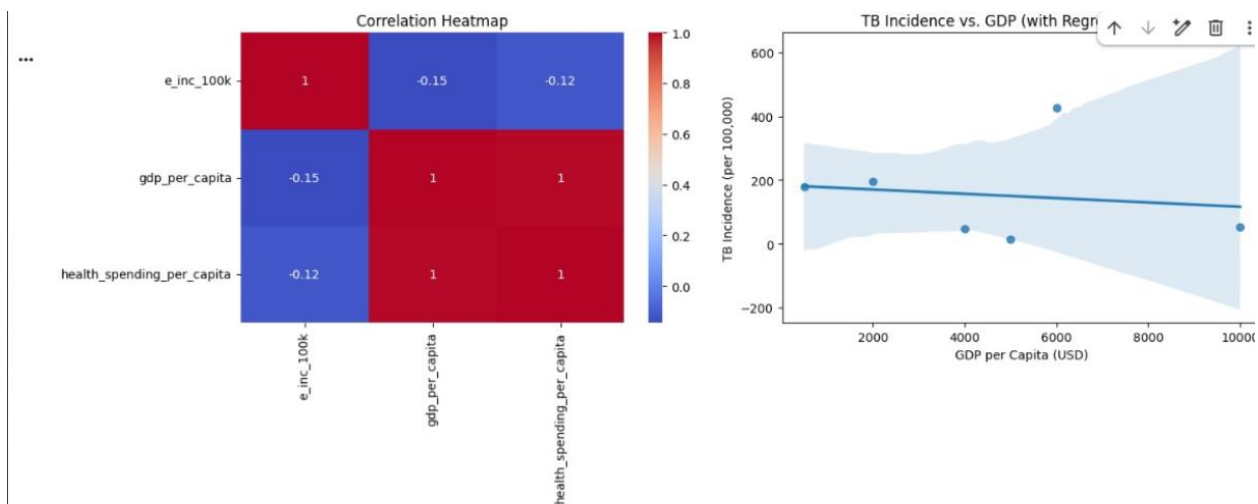
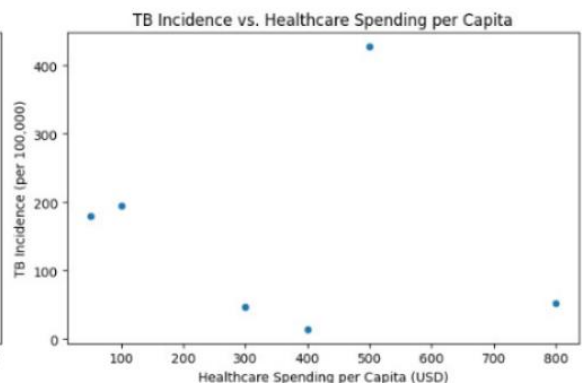
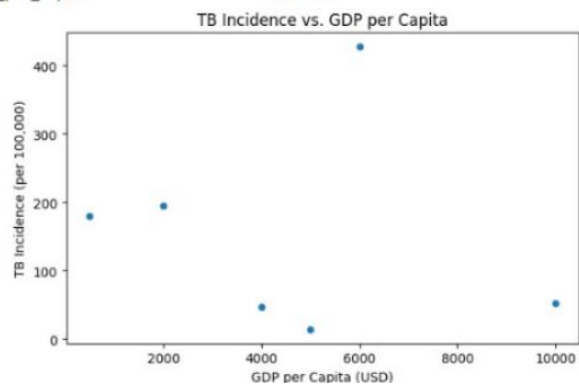
plt.tight_layout()
plt.show()

# Optional: Print key insights
print("\nKey Insights:")
```

Correlation Matrix:

	e_inc_100k	gdp_per_capita	\
e_inc_100k	1.000000	-0.146239	
gdp_per_capita	-0.146239	1.000000	
health_spending_per_capita	-0.115517	0.995208	

	health_spending_per_capita
e_inc_100k	-0.115517
gdp_per_capita	0.995208
health_spending_per_capita	1.000000



Key Insights:
Correlation between TB Incidence and GDP: -0.15
Correlation between TB Incidence and Healthcare Spending: -0.12
Countries analyzed: 6

Conclusion:

This project will provide an analytical view of tuberculosis trends worldwide. By leveraging WHO data and visualization tools, it will uncover valuable patterns to support global health decision-making and disease prevention strategies.

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